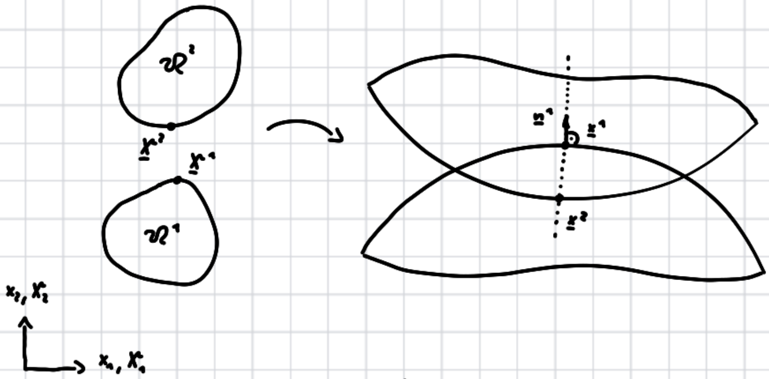
Contact Hertz



$$\underline{n}^1 \underline{x}^2 - \underline{n}^1 \underline{x}^2 = g_N \qquad g_N < 0 \leftarrow \text{Penetration}$$

$$(\underline{x}^2 - \underline{x}^1)\, \underline{n}^1 = g_N$$

$$g_N \geq 0$$

$$(\underline{x}^2 - \underline{x}^1)\, \underline{n}^1 \geq 0$$

$$\underline{x}^1 = \underline{u}^1 + \underline{X}^1$$

$$\underline{x}^2 = \underline{u}^2 + \underline{X}^2$$

Signori condition

$$g_N \geq 0 \quad p_N \geq 0 \quad p_N\, g_N = 0$$

$$\delta W_c = \int_{\Gamma_c} p_N\, \delta g_N\, d\Gamma$$

$$\delta g_N = (\delta \underline{x}^2 - \delta \underline{x}^1)\, \underline{n}^1$$

Small deformation

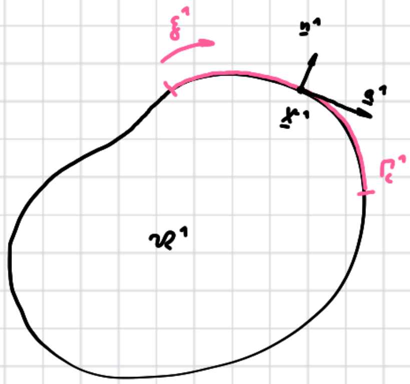
$$(\underline{u}^2 - \underline{u}^1)\, \underline{n}^1 + g_r = g_N \geq 0$$

$$g_r = (\underline{X}^2 - \underline{X}^1)\, \underline{n}^1$$

Penalty - Method

$$p_N = \begin{cases} 0 & g_N > 0 \\ \epsilon(-g_N) & g_N < 0 \end{cases}$$

$$\delta W_c = -\int_{\Gamma_c} p_N \, \delta g_N \, d\Gamma$$

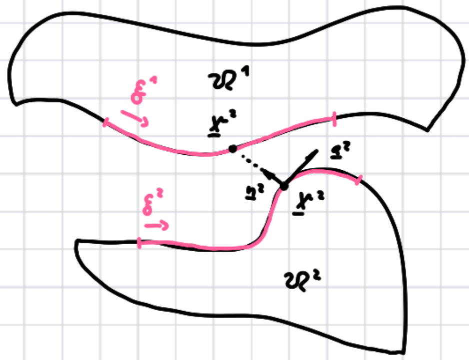


$$\underline{x}^1(\xi^1) = \sum_e N_e^1(\xi^1)\, \underline{x}_e^1$$

$$\underline{u}^1(\xi^1) = \sum_e N_e^1(\xi^1)\, \underline{u}_e^1$$

$$\underline{a}^1 = \frac{\partial \underline{x}^1}{\partial \xi^1} = \sum_e \frac{\partial N_e^1}{\partial \xi^1} \underline{x}_e^1$$

$$\underline{n}^1 = \frac{1}{\|\underline{a}^1\|} \begin{bmatrix} a_y^1 \\ a_x^1 \end{bmatrix}$$



search for ξ^2

$$(\underline{x}^2(\xi^2) - \underline{x}^1) \cdot \underline{a}^2 = 0$$

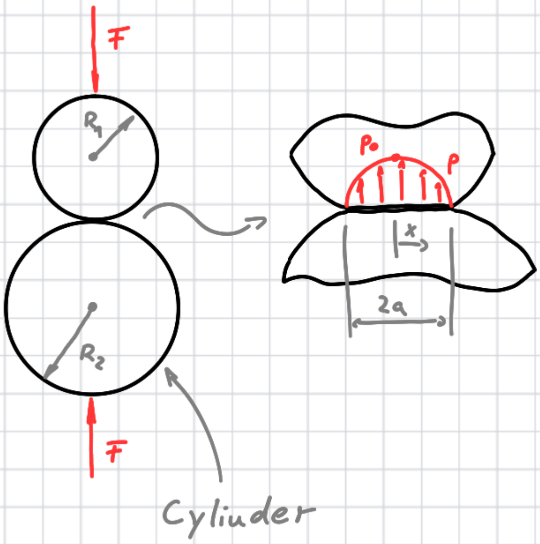
$$\underline{a}^2 = \frac{\partial \underline{x}^2}{\partial \xi^2} = \sum_e \frac{\partial N_e^2}{\partial \xi^2} \underline{x}_e^2$$

$$\underline{x}^2(\xi^2) = \sum_e N_e^2 \, \underline{x}_e^2$$

$$\underline{u}^2(\xi^2) = \sum_e N_e^2 \, \underline{u}_e^2$$

$$\delta g_N = (\delta \underline{u}^2 - \delta \underline{u}^1) \cdot \underline{n}^1 = \left(\sum_e N_e^2 \, \delta \underline{u}_e^2 - \sum_e N_e^1 \, \delta \underline{u}_e^1 \right) \cdot \underline{n}^1$$

Contact pressure



$$\frac{1}{R^*} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{E^*} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$$

$$p = p_0 \sqrt{1 - \frac{x^2}{a^2}} \qquad a = \sqrt{\frac{4FR^*}{\pi E^*}}$$

$$p_0 = \frac{2F}{\pi a}$$

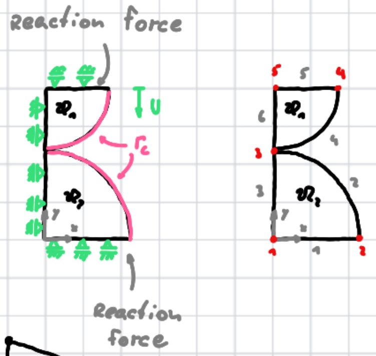
Plain strain

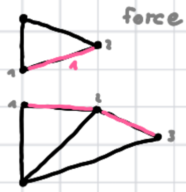
$$\mu = \frac{E}{2(1+\nu)} \qquad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

$R_1 = 12\,mm \qquad E_1 = 2{,}1 \cdot 10^5\ N/mm^2 \qquad \nu_1 = 0{,}3$

$R_2 = 15\,mm \qquad E_2 = 1{,}0 \cdot 10^5\ N/mm^2 \qquad \nu_2 = 0{,}25$

FEM





$$N_1 = 1 - \xi \qquad \partial N_1 / \partial \xi = -1$$

$$N_2 = \xi \qquad \partial N_2 / \partial \xi = 1$$

1 → search segment

2 set Gauss nodes

$$\underline{x}^1 = N_1^1(\xi^1)\,\underline{x}_1^1 + N_2^1(\xi^1)\,\underline{x}_2^1$$

$$\underline{u}^1 = N_1^1(\xi^1)\,\underline{u}_1^1 + N_2^1(\xi^1)\,\underline{u}_2^1$$

$$\underline{a}^1 = \frac{\partial \underline{x}^1}{\partial \xi^1}(\xi^1) = \frac{\partial N_1^1}{\partial \xi^1}(\xi^1)\,\underline{x}_1^1 + \frac{\partial N_2^1}{\partial \xi^1}(\xi^1)\,\underline{x}_2^1$$

$$\underline{n}^1 = \frac{1}{\|\underline{a}^1\|} \begin{bmatrix} a_y^1 \\ -a_x^1 \end{bmatrix}$$

$$\left(\underline{x}^2(\xi^2) - \underline{x}^1\right) \cdot \underline{a}^2 = 0$$

search ξ^2

$$(\underline{u}^2 - \underline{u}^1)\, \underline{n}^1 + g_X = g_N$$

$$g_X = \left(\underline{X}^2 - \underline{X}^1\right) \underline{n}^1$$

$$p_N = \begin{cases} 0 & g_N > 0 \\ \epsilon\,(-g_N) & g_N < 0 \end{cases}$$

$$\delta g_N = (\delta \underline{u}_2 - \delta \underline{u}_1) \cdot \underline{n}_1 = \left(\sum_e N_e^2\, \delta \underline{u}_e^2 - \sum_e N_e^1\, \delta \underline{u}_e^1 \right) \cdot \underline{n}^1$$

$$\int_{\Omega} \underline{\underline{\sigma}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v})\, dx - \int_{\Gamma_c} p_N\, \delta g_N\, d\Gamma = 0$$

$$\xi^2 = \left(\underline{x}^1 - \underline{x}_1^2 \right) \cdot \underline{a}^2 / (\underline{a}^2 \cdot \underline{a}^2)$$

$$\underline{a}_2 = \underline{X}_2^2 - \underline{X}_1^2$$

$$\int_{\Omega} \underline{\underline{\sigma}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v})\, dx - \int_{\Gamma_c} p_N\, \delta g_N\, d\Gamma = 0$$

$$\int_{\Omega} \underline{\underline{\sigma}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v})\, dx = \underline{v}^T \underline{\underline{K}}\, \underline{u}$$

$$-\int_{\Gamma_c} p_N\, \delta g_N\, d\Gamma = -\int_{\Gamma_c} \epsilon(-g_N)\, \delta g_N\, d\Gamma \qquad g_N = (\underline{u}^2 - \underline{u}^1)^T \underline{n}^1 + g_x$$

$$\delta g_N = (\delta \underline{u}^2 - \delta \underline{u}^1)^T \underline{n}^1$$

$$= \epsilon \int_{\Gamma_c} g_N\, \delta g_N\, d\Gamma$$

$$\underline{u}^1 = N_1^1 \underline{u}_1^1 + N_2^1 \underline{u}_2^1$$

$$g_N = \underline{n}^{1T}(\underline{\underline{N}}^2 - \underline{\underline{N}}^1)\underline{u} + g_x$$

$$\underline{u}^1 = \begin{bmatrix} N_1^1 & 0 & N_2^1 & 0 & \dots \\ 0 & N_1^1 & 0 & N_2^1 & \end{bmatrix} \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ \vdots \end{bmatrix}$$

$$g_N = \underline{c}^T \underline{u} + g_x$$

$$\delta g_N = \underline{n}^{1T}(\underline{\underline{N}}^2 - \underline{\underline{N}}^1)\underline{v}$$

$$\underline{u}^2 = \begin{bmatrix} \dots & N_1^2 & 0 & N_2^2 & 0 \\ & 0 & N_1^2 & 0 & N_2^2 \end{bmatrix} \begin{bmatrix} \vdots \\ u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \end{bmatrix}$$

$$\delta g_N = \underline{c}^T \underline{v} = \underline{v}^T \underline{c}$$

$$= \epsilon\, \underline{v}^T \int_{\Gamma_c} \left[\underline{c}\,\underline{c}^T \underline{u} + \underline{c}\, g_x\right] d\Gamma \qquad \underline{c} = (\underline{\underline{N}}^2 - \underline{\underline{N}}^1)^T \underline{n}^1$$

$$\underline{u}^1 = \underline{\underline{N}}^1 \underline{u} \qquad \delta \underline{u}^1 = \underline{\underline{N}}^1 \underline{v}$$

$$= \underline{v}^T \underline{\underline{K}}_c\, \underline{u} + \underline{v}^T \int_{\Gamma_c} \epsilon\, \underline{c}\, g_x\, d\Gamma \qquad \underline{u}^2 = \underline{\underline{N}}^2 \underline{u} \qquad \delta \underline{u}^2 = \underline{\underline{N}}^2 \underline{v}$$

$$\underline{\underline{K}}_c = \epsilon \int_{\Gamma_c} \underline{c}\,\underline{c}^T d\Gamma = \epsilon \|a_1\| \sum (w_1\, \underline{c}\,\underline{c}^T + w_2\, \underline{c}\,\underline{c}^T)$$

$$d\Gamma = \|a_1\|\, d\xi \qquad w_1 = 0{,}5 \quad \xi_1 = 1/2 + 1/2 \frac{1}{\sqrt{3}}$$

$$w_2 = 0{,}5 \quad \xi_2 = 1/2 - 1/2 \frac{1}{\sqrt{3}}$$

Newton: insert u_D

Only active if $g_N = \underline{c}^T \underline{u} + g_x < 0$

$$\underline{u}^{N+1} = \underline{u}^N + \Delta \underline{u} \qquad \underline{\underline{J}} = \underline{\underline{K}} + \underline{\underline{K}}_c$$

$$\underline{\underline{J}}(\underline{u}^N)\Delta \underline{u} = -\underline{R}(\underline{u}^N) \qquad \underline{R} = \underline{\underline{K}}\,\underline{u}^N + \epsilon \int_{\Gamma_c} \left[\underline{c}\,\underline{c}^T \underline{u}^N + \underline{c}\, g_x\right] d\Gamma$$

$\Delta u_i = 0$ where u_D is set